

CHUNGGANG, Korea, Democratic People's Republic ofWMO: **470140**Lat: **41.783N**Lon: **126.883E**Elev: **331**StdP: **97.41**Time Zone: **9.00 (E09)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.0	-25.0	-28.9	0.3	-26.0	-27.2	0.3	-24.4	5.5	-4.8	4.4	-7.6	0.0	N/A	0.423

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.9	31.4	24.2	29.9	23.8	28.5	22.7	26.2	30.0	24.9	28.5	23.8	27.2	1.3	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.1	21.0	28.9	23.8	19.4	27.6	22.7	18.2	26.3	83.7	30.4	78.2	28.6	73.6	27.2	32.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.5	4.4	3.4	-29.9	34.0	1.8	1.4	-31.1	35.0	-32.2	35.8	-33.2	36.6	-34.5	37.7		
(5)			-30.0	27.7	1.8	2.1	-31.2	29.3	-32.3	30.5	-33.3	31.7	-34.5	33.3	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.1	-15.6	-10.7	-1.1	8.1	15.1	20.1	23.6	22.6	16.6	8.2	-2.0	-12.5	(6)	
	DBStd	14.12	4.43	5.36	5.08	4.25	3.46	2.69	2.28	2.78	3.43	3.85	5.44	4.98	(7)	
	HDD10.0	2943	792	578	344	84	4	0	0	0	2	80	361	697	(8)	
	HDD18.3	4849	1051	812	602	308	109	13	1	4	72	313	610	956	(9)	
	CDD10.0	1528	0	0	0	27	162	304	422	391	198	25	0	0	(10)	
	CDD18.3	394	0	0	0	0	8	67	164	136	18	0	0	0	(11)	
	CDH23.3	2747	0	0	0	16	184	530	1087	819	109	3	0	0	(12)	
	CDH26.7	797	0	0	0	1	40	152	355	239	10	0	0	0	(13)	
(14) Wind	WSAvg	0.6	0.3	0.4	0.7	1.1	1.1	0.7	0.4	0.4	0.5	0.6	0.6	0.3	(14)	
(15) Precipitation	PrecAvg	837	10	11	17	51	75	120	203	195	76	36	30	15	(15)	
	PrecMax	1238	27	33	40	132	168	226	318	416	152	82	66	28	(16)	
	PrecMin	644	2	1	4	8	40	37	77	89	18	5	6	4	(17)	
	PrecStd	135	7	8	10	28	31	46	72	85	37	21	16	6	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.1	5.9	15.7	25.8	30.4	32.2	34.1	33.1	28.1	23.2	13.6	2.1	(19)	
		MCWB	0.0	2.5	10.1	15.8	19.8	22.0	26.8	27.0	20.9	15.6	8.7	0.0	(20)	
	2%	DB	-2.7	2.4	12.4	22.1	27.6	30.1	31.7	30.9	26.1	20.1	10.4	-0.4	(21)	
		MCWB	-4.3	0.2	7.3	13.5	18.0	21.7	25.3	25.1	19.9	14.0	7.1	-1.6	(22)	
	5%	DB	-5.3	0.7	9.3	19.2	25.3	28.4	30.1	29.2	24.5	17.8	7.8	-2.6	(23)	
		MCWB	-6.5	-1.1	5.0	12.1	17.0	20.8	24.7	24.1	18.8	12.1	5.4	-3.8	(24)	
	10%	DB	-7.9	-1.1	6.6	16.8	23.0	26.7	28.6	27.7	22.7	15.4	5.5	-4.4	(25)	
		MCWB	-9.0	-2.8	3.3	10.3	15.9	20.4	23.9	23.2	17.5	10.7	3.5	-5.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.2	2.7	10.6	17.4	21.9	24.5	28.4	28.6	22.7	16.4	9.5	0.3	(27)	
		MCDB	1.4	5.2	15.0	23.4	27.1	28.8	32.3	30.9	25.9	21.7	12.5	1.6	(28)	
	2%	WB	-4.3	0.5	7.6	14.6	19.6	23.0	26.6	26.1	21.0	14.3	7.5	-1.6	(29)	
		MCDB	-2.9	2.1	11.8	20.3	25.5	28.1	30.5	29.7	24.4	19.1	10.1	-0.5	(30)	
	5%	WB	-6.6	-1.1	5.5	12.7	18.0	21.8	25.3	24.9	19.7	12.7	5.6	-3.7	(31)	
		MCDB	-5.4	0.5	8.8	18.3	23.8	26.9	29.1	28.3	23.3	17.0	7.5	-2.6	(32)	
	10%	WB	-9.0	-2.8	3.6	10.9	16.4	20.7	24.3	23.8	18.5	11.1	3.7	-5.4	(33)	
		MCDB	-8.0	-1.2	6.3	16.0	21.8	25.5	27.7	26.9	21.8	14.7	5.3	-4.4	(34)	
(35) Mean Daily Temperature Range	MDBR		12.9	13.8	11.6	12.8	13.1	11.0	8.9	9.2	11.8	12.7	9.5	11.0	(35)	
	5% DB	MCDBR	13.8	13.4	14.5	18.2	17.8	14.7	11.6	11.1	13.5	16.2	11.9	11.4	(36)	
		MCWBR	12.4	11.3	10.2	11.1	9.5	7.5	6.7	6.4	7.9	10.3	9.0	10.1	(37)	
	5% WB	MCDBR	13.7	13.0	13.8	16.5	15.9	12.6	10.4	9.8	11.4	14.8	11.2	11.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.360	0.425	0.501	0.574	0.616	0.646	0.591	0.523	0.465	0.485	0.427	0.372	(40)	
	taud		1.979	1.833	1.749	1.687	1.647	1.658	1.838	2.005	2.074	1.909	1.948	1.981	(41)	
	Ebn at Noon		735	738	729	711	695	676	710	748	762	671	648	676	(42)	
	Edh at Noon		121	165	203	232	248	245	203	167	145	152	126	112	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.41	3.32	4.34	4.74	5.33	5.16	4.63	4.60	4.29	3.22	2.12	1.92	(44)	
	RadStd		0.15	0.17	0.22	0.33	0.47	0.53	0.48	0.45	0.40	0.26	0.26	0.14	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page